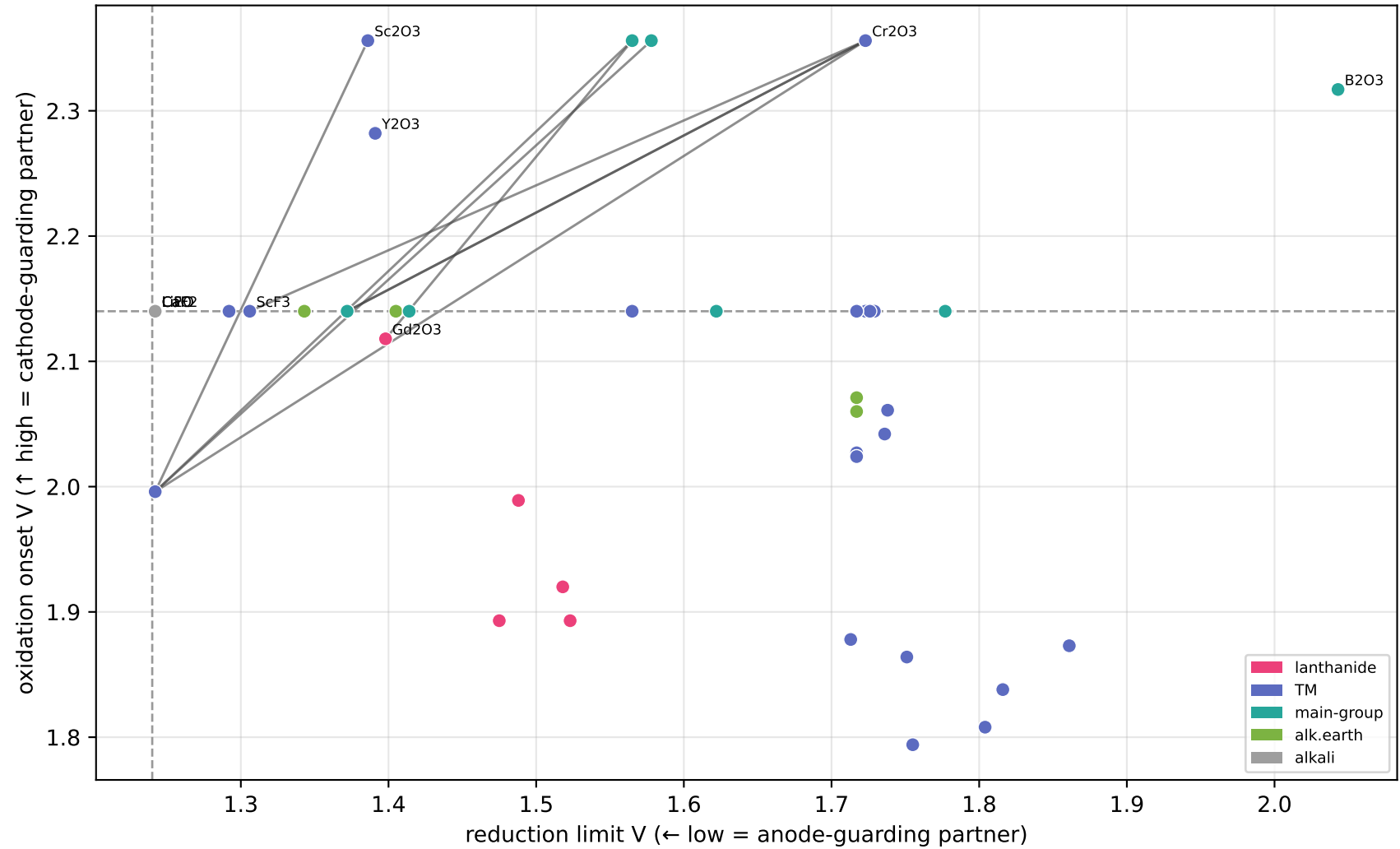
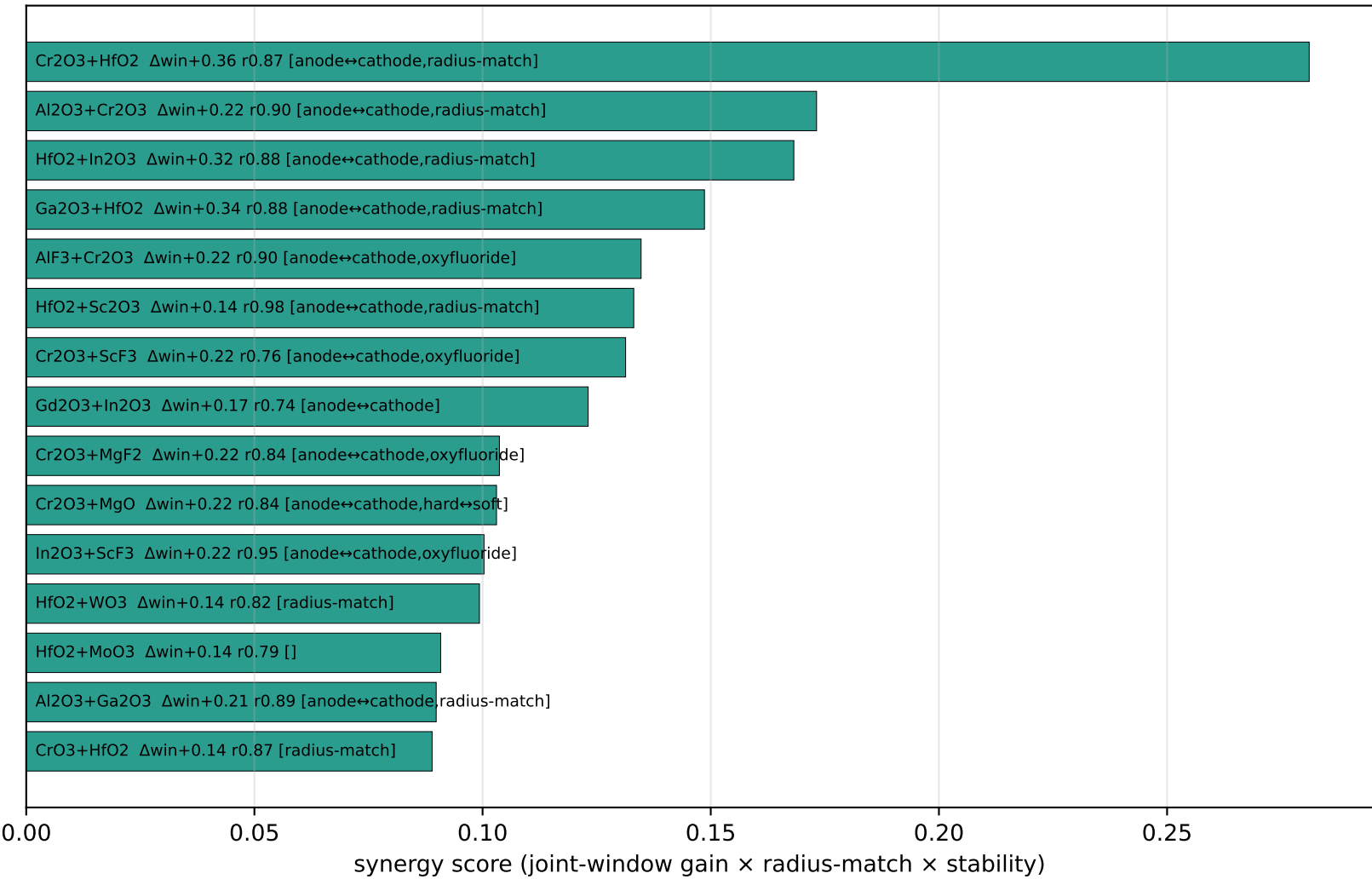


Cascade v23 — CO-DOPING SYNERGY hypotheses (single-dopant proxy; NOT validated — needs co-doped DFT)

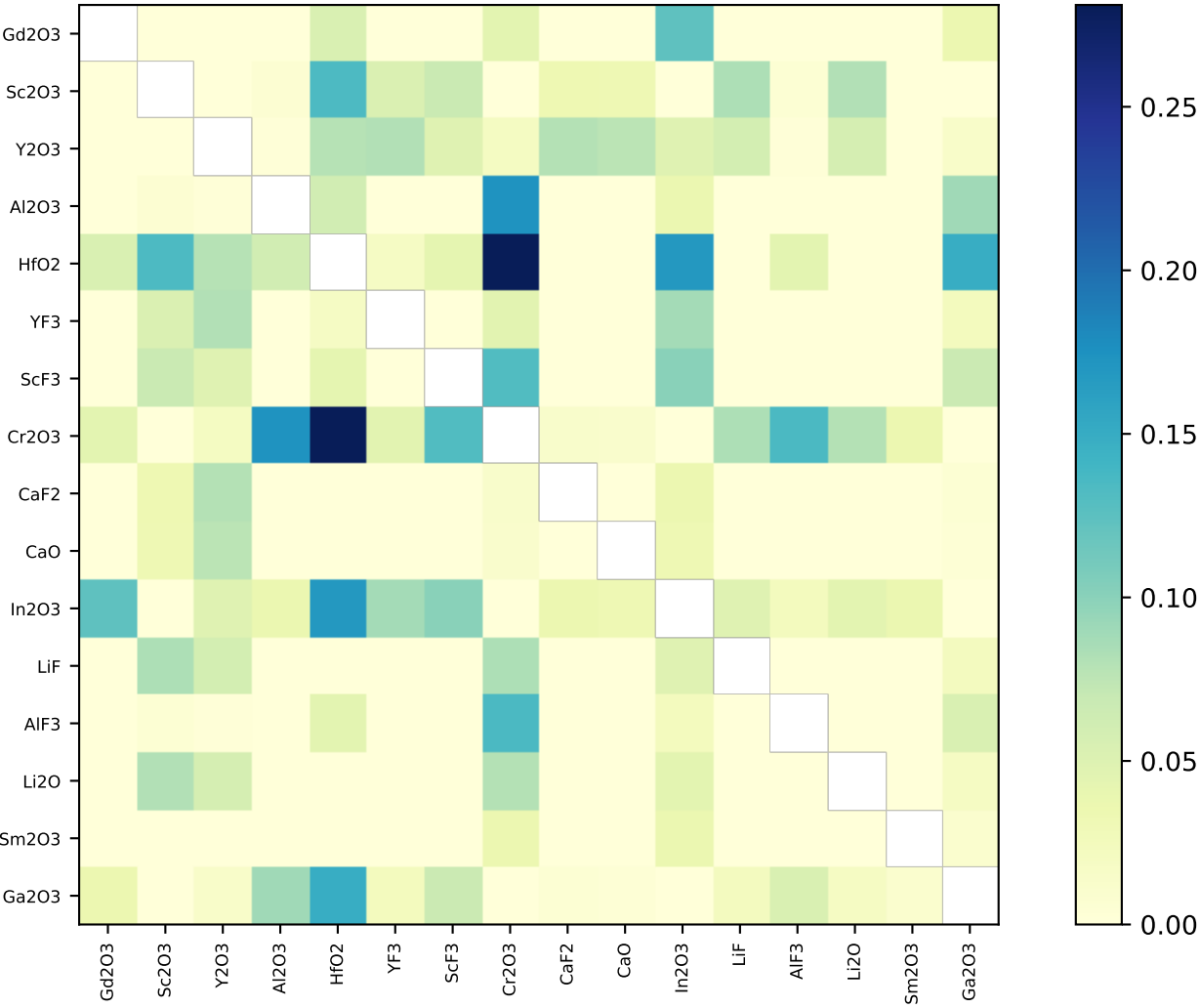
(A) Anode↔cathode complementarity — lines = top-8 synergy pairs
ideal pair spans low-red + high-ox



(B) Top-15 co-doping synergy candidates



(C) Pairwise synergy among 16 best single dopants



(D) Oxyfluoride co-doping (oxide bulk-stability + fluoride LiF-SEI)

